



Framework for probabilistic tsunami hazard assessment considering the effects of sea-level rise due to climate change

Abdul Kadir Alhamid^a, Mitsuyoshi Akiyama^{a,*}, Hiroki Ishibashi^b, Koki Aoki^a, Shunichi Koshimura^c, Dan M. Frangopol^d

^a Department of Civil and Environmental Engineering, Waseda University, Tokyo 169-8555, Japan

^b Department of Civil Engineering, College of Engineering, Nihon University, Koriyama 963-8642, Japan

^c International Research Institute of Disaster Science, Tohoku University, Sendai 980-0845, Japan

^d Department of Civil and Environmental Engineering, Engineering Research Center for Advanced Technology for Large Structural Systems (ATLSS Center), Lehigh University, Bethlehem, PA 18015-4729, USA

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ABSTRACT

Sea-level rise due to climate change could significantly exacerbate tsunami disasters since the sea level is a critical parameter affecting the intensity of tsunamis. Considering the impacts of future climate change on the ocean, a method to consider the effects of sea-level rise on the tsunami hazard intensity is needed to precisely predict future tsunami disasters. This paper presents a novel framework for probabilistic tsunami hazard assessment, considering the sea-level rise and associated uncertainties. A probabilistic assessment of the sea-level rise hazard is performed using the available data and climate models considering several climate change emission scenarios. Conditional tsunami hazard curves are estimated by conducting tsunami propagation simulations given a sea-level rise while considering the uncertainties associated with fault movement (i.e., rake angles and average stress drops). Radial-basis-function-based surrogate models and quasi-Monte-Carlo simulations are employed to obtain tsunami hazard curves. Finally, the tsunami hazard curves considering the effects of sea-level rise are estimated by convolving the corresponding regional sea-level rise hazard with the conditional tsunami hazard curves based on the total probability theorem. An illustrative example is provided, in which the proposed framework is applied to several municipalities in the Mie Prefecture of Japan that would be affected by tsunamis during the anticipated Nankai-Tonankai earthquake. The effects of sea-level rise on the tsunami hazard intensities associated with the municipalities are discussed.

1. Introduction

Tsunamis have been widely recognized as devastating natural hazards, leading to significant losses of coastal infrastructure assets and casualties [1–3]. For example, during the 2011 Great East Japan earthquake, a destructive tsunami caused damage to or washed away more than 200 bridges and generated approximately 23 million tons of disaster waste [4,5]. Another example is the notorious 2004 Sumatra-Andaman tsunami, which eradicated coastal infrastructure and caused major disruption to 3,000 km of road segments in Banda Aceh in Indonesia [6,7]. Disaster preparedness for future tsunamis must be enhanced to increase the resilience of communities in coastal regions [8,9].

Tsunami hazard assessment has been extensively studied to quantify the potential impact of a tsunami and is considered a valuable disaster-risk mitigation tool for decision-makers [10–13]. Considerable efforts have been devoted to alleviate tsunami-induced disasters, including the estimation of cascading earthquake-tsunami impacts on building losses [14,15], assessment of road network risk and resilience [16,17], and estimation of the generated disaster waste [18]. However, tsunami disasters have been evaluated as time-independent events, even though climate change is causing a gradual sea-level rise. Therefore, tsunami hazard assessment methodologies must be further developed by considering multiple events and time-dependent hazards to accurately evaluate the reliability, risk, and resilience of structures and infrastructure systems [19].

* Corresponding author at: Department of Civil and Environmental Engineering, Waseda University, Building 51-16-09, 3-4-1 Okubo Shinjuku-Ku, Tokyo 169-8555, Japan.

E-mail address: akiyama617@waseda.jp (M. Akiyama).

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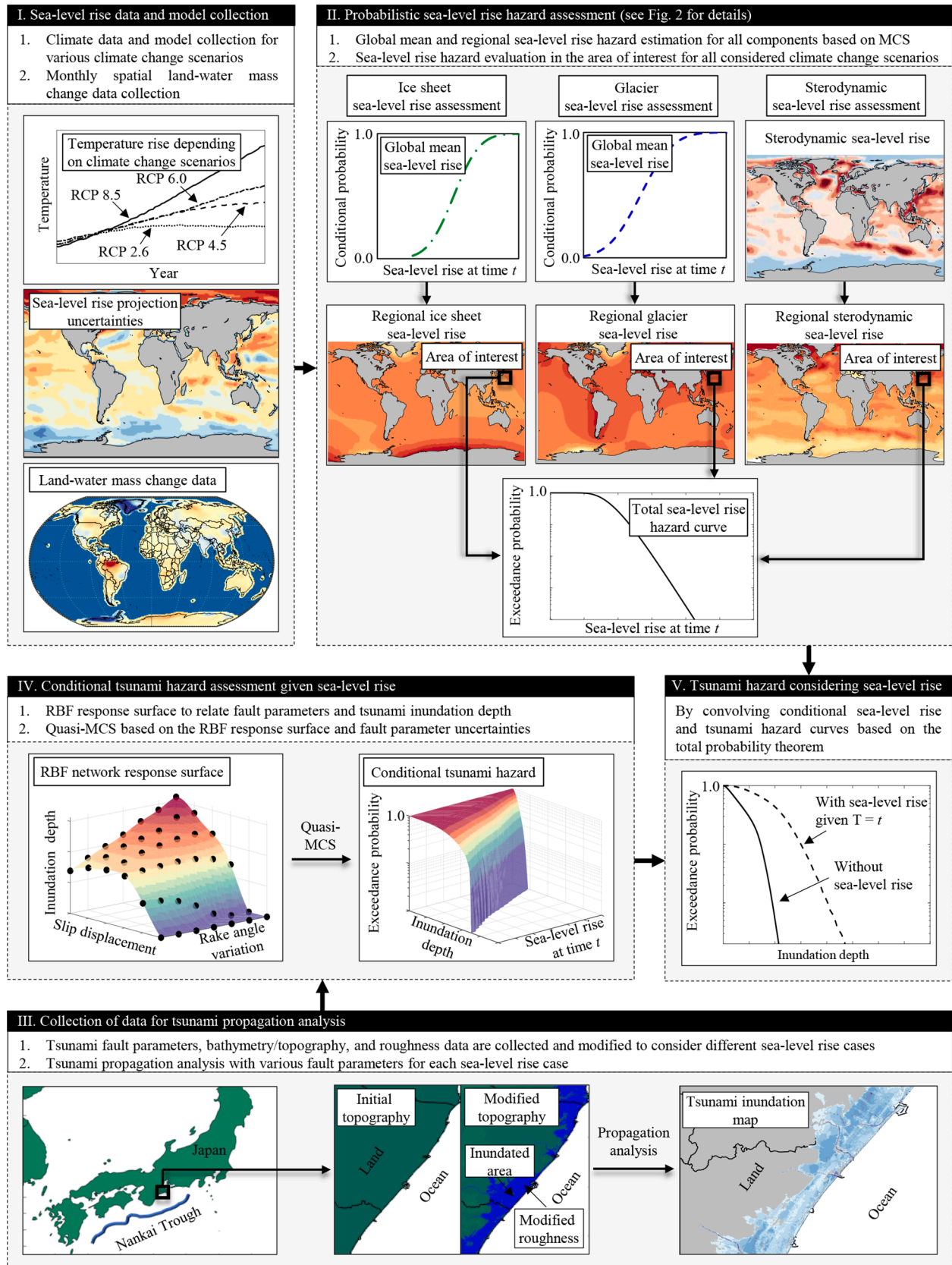


Fig. 1. Procedure for estimating tsunami hazards considering the effects of sea-level rise.

Anthropogenic greenhouse gas emissions have been rising during the past two decades and have accelerated the global warming process [20–22]. One primary climate issue addressed by the Intergovernmental Panel on Climate Change Fifth Assessment Report (IPCC AR5) is future global-scale sea-level rise, which is a risk to hundreds of millions of people who reside in low-elevation coastal areas that are vulnerable to permanent flooding [23,24]. Therefore, regional sea-level rise assessment is an imperative task to anticipate future coastal infrastructure losses. Several studies have comprehensively examined the available probabilistic sea-level rise approaches, such as process-based projections that exclusively estimate each physical process generating the sea-level rise and semiempirical assessment involving readjustment of the climate model to past sea-level data [24–27]. However, these studies primarily focused on probabilistic sea-level rise assessments relative to the past and assumed mutually exclusive processes among the sea-level rise components.

Furthermore, considerable research has been performed on climate change impacts across various sectors, including periodic storm surges [28,29], tropical cyclones [30–32], corrosion rate acceleration on exposed structures [33–37], bridge scour risk [38–43], local flooding due to extreme precipitation [44], and road infrastructure [45–50], as well as on modeling practices to quantify the influence of climate change [51–53]. The irreversible sea-level rise process may inherently threaten low-lying coastal infrastructure systems [54–56]. However, limited research has been devoted to investigating sea-level rise effects in tsunami hazard assessments. A recent numerical study demonstrated the effects of an increase of 0.5 m and 1 m in the sea level on the tsunami hazards in the Macau Special Administrative Region and clarified that an increase of 0.5 m in the sea level could double the tsunami hazard [57]. In addition, several studies have confirmed the contribution of sea-level rise to the intensification of tsunami impacts [58,59]. Nevertheless, the existing studies adopted deterministic sea-level rise scenarios without considering the aleatoric and epistemic uncertainties [52,60,61]. It is necessary to consider the uncertainties associated with sea-level rise prediction to avoid overestimating the reliability of structures under tsunami hazards. The integrated probabilistic approach of sea-level rise in tsunami hazard assessments can facilitate a more accurate risk evaluation of structures and enhance the resilience of coastal communities.

This paper presents a methodology for tsunami hazard assessment considering the effects of sea-level rise due to climate change. Probabilistic sea-level rise hazard assessment is performed by utilizing climate models and available data regarding sea-level rise components, including the uncertainties associated with possible climate change emission scenarios. Tsunami propagation analysis is performed considering the uncertainty associated with the prediction of earthquake fault movements (i.e., rake angles and average stress drops) given an increase in the sea level. A radial basis function (RBF) metamodel representing the relationship between the fault parameters and tsunami intensity is constructed using an experimental design method to reduce the computational time for tsunami hazard assessment [62]. Monte Carlo simulations (MCSs) are performed using a low discrepancy sequence (i.e., quasi-Monte-Carlo [63]) to generate conditional tsunami hazard curves given sea-level rise. Finally, based on the total probability theorem, tsunami hazard curves considering the effects of sea-level rise are obtained. The proposed framework is implemented in a case study involving several municipalities in Mie Prefecture of Japan, in which the tsunami intensity due to the anticipated Nankai-Tonankai earthquake is expected to be intense. The effects of the sea-level rise on the tsunami intensity for each considered city are discussed.

2. Basic procedure for estimating tsunami hazards considering the effects of sea-level rise due to climate change

Figure 1 shows the procedure to estimate tsunami hazards considering the effects of sea-level rise due to climate change. The proposed

Table 1

Uncertainties associated with sea-level rise hazard assessment.

Sea-level rise components	Input variables	Associated uncertainties	References
Sterodynamic sea-level rise	Thermosteric sea-level rise STR	Climate change emission scenarios Π_j General circulation models m	[70]
	Dynamic sea-level rise ΔDSL	Climate change emission scenarios Π_j General circulation models m	[70]
Glacier sea-level rise	Probabilistic global mean glacier sea-level rise $G(t)$	Probabilistic projection from several glacier models	[71]
	Glacier sea-level fingerprints FG_{GLA}	Monthly data measured from 2006 to 2016	[83]
Ice sheet sea-level rise	Global mean ocean surface temperature $\Delta\tau$	Climate change emission scenarios Π_j General circulation models m	[70]
	Global mean surface temperature $\Delta\tau_a$	Climate change emission scenarios Π_j General circulation models m	[70]
	Volume above floatation change ΔVAF	Ice sheet models l	[72]
	Subshelf ice discharge B	Ice sheet models l	[72]
	Basal melt sensitivity β	Uniform distribution	[85]
	Glacier sea-level fingerprints FG_{ICS}	Monthly data measured from 2006 to 2016	[83]

method consists of five main steps: (I) sea-level rise data and climate model collection, (II) probabilistic sea-level rise hazard assessment, (III) tsunami data collection and tsunami inundation analysis, (IV) conditional tsunami hazard assessment given sea-level rise, and (V) estimation of the tsunami hazard curve considering the effects of sea-level rise. As shown in Fig. 1, the tsunami hazard curve that includes the effects of sea-level rise is evaluated by convolving the corresponding sea-level rise hazard in an area of interest (i.e., tsunami fault location) and conditional tsunami hazard curves based on the total probability theorem. To perform the tsunami hazard assessment considering the effects of sea-level rise due to climate change, the following assumptions are made:

- [A] Epistemic uncertainties due to the varying data output among climate models and disagreement among experts' projections are considered the main uncertainties in identifying sea-level rise hazards, as listed in Table 1. Other uncertainties associated with the geophysical process of sea-level rise, including its numerical modeling, are beyond the scope of this study.
- [B] Common sea-level rise projections are conditionally dependent on plausible future climate change emission scenarios, which are known as the representative concentration pathways (RCPs) and defined as emissions from energy and land-use trends [64,65]. In this paper, four different emission scenarios (i.e., RCP 2.6, RCP 4.5, RCP 6.0, and RCP 8.5) are considered in the sea-level rise hazard assessment. Since no consensus exists regarding the likelihood of each future climate change emission scenario [61,66], the occurrence probabilities associated with each scenario are assumed to be identical.
- [C] As sea-level rise components due to climate change, sterodynamic, glacier, and ice sheet sea-level rise are considered. Other sea-level rise components, such as land water storage from hydrological processes and inland water utilization, are disregarded

since the sea-level rise contributions of these components are negligible [66].

- [D] Tsunami hazards given the occurrence of an earthquake are estimated considering the uncertainties associated with the fault movement parameters. Since the short-term effects of sea-level rise due to climate change are negligible, a tsunami hazard curve is estimated under the condition that the earthquake occurs at a specific time (e.g., in 100 years). The probabilistic tsunami hazard intensity and sea-level rise at that time are integrated.
- [E] Nonstationary processes such as land subsidence [67] are not considered. Further research is needed to conduct probabilistic hazard assessments in the presence of nonstationary action [68].
- [F] Since detailed information regarding the coastal structures along the coastline is not available, tsunami propagation analysis is conducted without considering their presence.

To estimate the sea-level rise hazard, preliminary data and the uncertainties associated with the considered sea-level rise components and climate models are obtained from various sources and references in step (I). The data include ocean and land surface temperature changes in different climate change emission scenarios, various sea-level rise projections from several climate models, and land-water mass change measurements from satellite gravimetry. Subsequently, a probabilistic sea-level rise hazard assessment is performed in step (II). As illustrated in Fig. 1, the probabilistic global mean sea-level rise (i.e., the average value of sea-level rise over the ocean) for the three sea-level rise components is calculated based on the climate data and uncertainties determined in step (I). In addition, the regional sea-level rise hazard (i.e., the sea-level rise value at any specific location in the ocean) is evaluated from the obtained global mean sea-level rise and land-water mass change data based on the MCS. The sea-level rise hazard over an earthquake fault is estimated considering the four climate change emission scenarios, as discussed in Section 3.

In step (III), bathymetry and topography data for tsunami propagation analysis are collected to account for different sea-level rise cases. When the land elevation lies below sea level, as demonstrated in Fig. 1, the roughness coefficient is modified and assumed identical to the ocean realm. In addition, to perform the probabilistic tsunami propagation analysis, the uncertainty associated with fault movement (i.e., rake angles and average stress drops) is quantified. In step (IV), to reduce the computational cost associated with the tsunami hazard assessment, the RBF network response surface is developed to relate the fault parameters with the corresponding tsunami intensity based on a limited number of tsunami propagation analyses. In this study, the inundation depth is used to reflect the tsunami intensity. Conditional tsunami hazard curves are estimated using the RBF network response surface with quasi-MCS, given the sea-level rise intensity.

Finally, in step (V), the tsunami hazard is obtained by convolving the conditional tsunami hazard curves with the corresponding sea-level rise hazard over the target earthquake fault location x_r in the ocean by using the total probability theorem:

$$P(H_n(x_r, t) > h) = \int_0^\infty \left\{ P(H_n(t) > h | \Psi = \psi(x_r, t)) \cdot \left| -\frac{dF_\Psi(\psi(x_r, t))}{d\psi} \right| \right\} d\psi \quad (1)$$

where $P(H_n(x_r, t) > h)$ is the tsunami hazard representing the probability of tsunami intensity $H_n(x_r, t)$ in the analyzed area n exceeding a value h at time t considering the effects of sea-level rise over the target earthquake fault location x_r , t is the specified time in years relative to the present, $P(H_n(t) > h | \Psi = \psi(x_r, t))$ is the conditional tsunami hazard curve given the sea-level rise intensity Ψ over the target earthquake fault location x_r equals $\psi(x_r, t)$, and $F_\Psi(\psi(x_r, t))$ is the cumulative distribution function (CDF) of the sea-level rise when the sea-level rise intensity Ψ is less than $\psi(x_r, t)$. In this study, the term sea-level rise hazard is defined as the probability of the complementary event (i.e., Ψ is at least $\psi(x_r, t)$).

An illustrative example of the proposed framework is provided in Section 5 to investigate the effects of the sea-level rise on the tsunami hazards associated with municipalities in Mie Prefecture of Japan.

3. Probabilistic sea-level rise hazard analysis

Given the RCP scenario Π_j , where Π_1 , Π_2 , Π_3 , and Π_4 correspond to RCP 2.6, RCP 4.5, RCP 6.0, and RCP 8.5, respectively, a conventional approach for estimating the regional sea-level rise is to multiply each global mean sea-level rise component by its associated sea-level spatial variability (defined as the sea-level fingerprint) and combining them as [69]:

$$\psi(x_r, t) = \omega(x_r, t) + \xi(x_r, t) + \eta(x_r, t) \quad (2)$$

$$\omega(x_r, t) = FG_{STR}(x_r) \cdot S(x_r, t) \quad (3)$$

$$\xi(x_r, t) = FG_{GLA}(x_r) \cdot G(t) \quad (4)$$

$$\eta(x_r, t) = FG_{ICS}(x_r) \cdot I(t) \quad (5)$$

where $\psi(x_r, t)$, $\omega(x_r, t)$, $\xi(x_r, t)$, and $\eta(x_r, t)$ are the total, sterodynamic, glacier, and ice sheet regional sea-level rise values at time t , respectively; $FG_{STR}(x_r)$, $FG_{GLA}(x_r)$, and $FG_{ICS}(x_r)$ are the sea-level fingerprints for the sterodynamic, glacier, and ice-sheet sea-level rise, respectively; $G(t)$ and $I(t)$ are the global mean glacier and ice sheet sea-level rise values, respectively; and $S(x_r, t)$ is the sterodynamic sea-level rise from global mean thermosteric and regional ocean dynamic sea-level rise. The sea-level fingerprints represent the expected sea-level rise per meter of global mean sea-level rise and are assumed to be time-independent according to previous studies [e.g., 27, 66, 69].

The probability distribution function (PDF) of each regional sea-level rise component depends on the associated uncertainties. The PDF of the sterodynamic sea-level rise is estimated through a conventional statistical analysis by utilizing various grid-based numerical climate model outputs known as general circulation models (GCMs). These models have been developed by 20 climate model groups worldwide, with more than 30 GCMs generated and collectively archived in the Coupled Model Intercomparison Project 5 (CMIP5) [70]. In terms of the glacier sea-level rise, the available glacier models involved in the Glacier Model Intercomparison Project (GlacierMIP) can be utilized for PDF estimation [e.g., 71]. In addition, MCSs can be performed to evaluate the PDF of the ice sheet sea-level rise based on the ice sheet dynamics and surface mass balance data from the Sea-Level Response to Ice Sheet Evolution (Sea-RISE) experiment [72].

It is assumed that the sterodynamic and glacier sea-level rise exhibit a normal distribution [73], the ice sheet sea-level rise exhibits a lognormal distribution [74], and the sea-level rise components are perfectly correlated with one another since they increase simultaneously over time with respect to global warming [25,26]. The conditional PDF of the total sea-level rise Ψ_j given any climate change emission scenario Π_j ($j = 1, 2, 3, 4$) can be derived from the convolution of the multiple probability distribution concept as:

$$f_{\Psi_j}(\psi(x_r, t) | \Pi_j) = \int_0^\infty \int_{-\infty}^\infty f_{SGI}(\omega(x_r, t), \xi(x_r, t), \eta(x_r, t) | \Pi_j) d\xi d\eta \quad (6)$$

where $f_{\Psi_j}(\psi(x_r, t) | \Pi_j)$ is the conditional PDF of the regional total sea-level rise Ψ_j given Π_j , and $f_{SGI}(\omega(x_r, t), \xi(x_r, t), \eta(x_r, t) | \Pi_j)$ represents the multivariate distribution pertaining to the sterodynamic, glacier, and ice sheet sea-level rise values, with regional sea-level rise values of $\omega(x_r, t)$, $\xi(x_r, t)$, and $\eta(x_r, t)$, respectively, given Π_j .

The total sea-level rise PDF and CDF over the target earthquake fault location x_r are calculated by incorporating the occurrence probability of climate change emission scenarios according to the total probability theorem. Based on the obtained total sea-level rise PDF, the CDF of the total sea-level rise can be determined as:

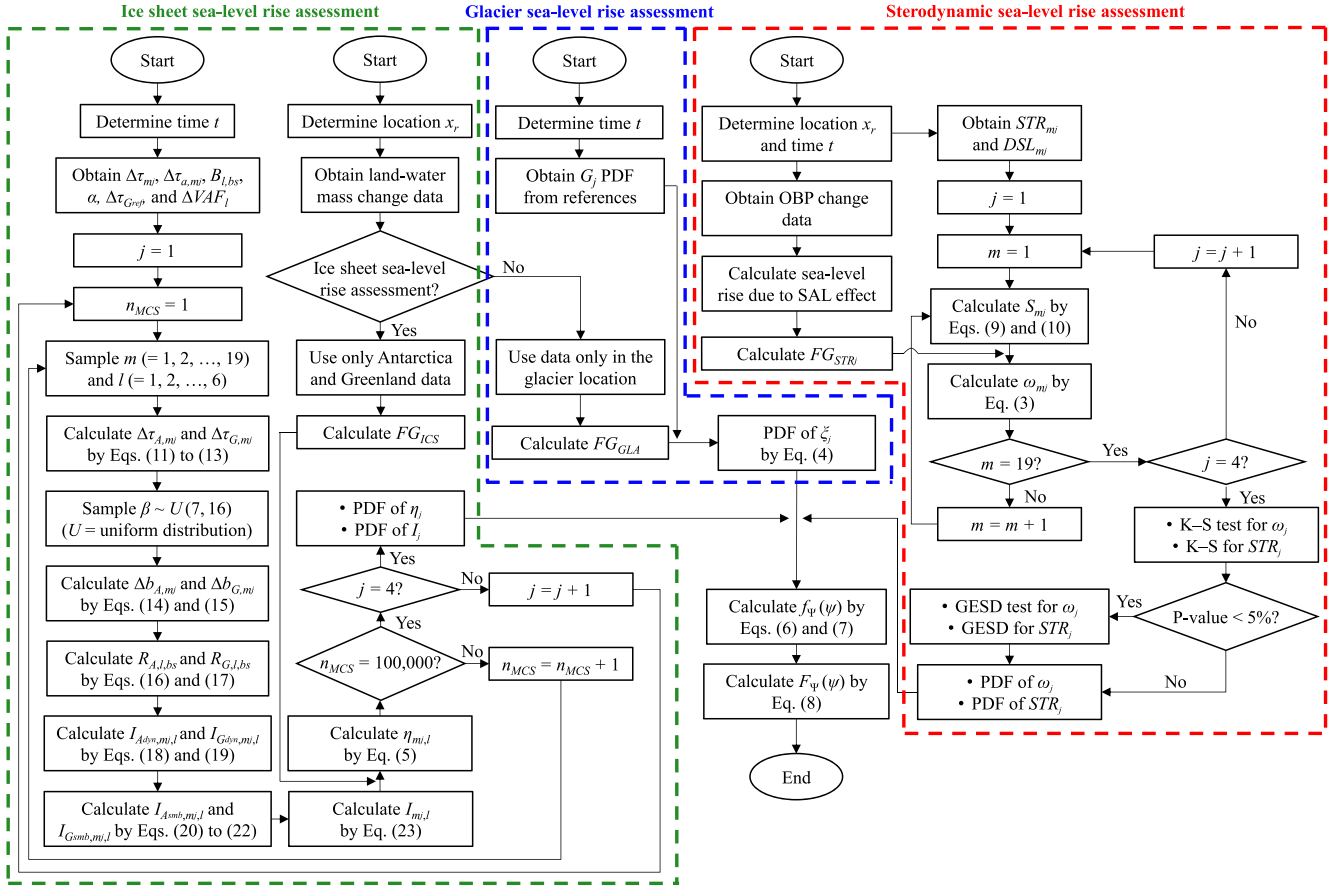


Fig. 2. Flowchart for probabilistic sea-level rise hazard assessment.

$$f_{\Psi}(\psi(x_r, t)) = \sum_{j=1}^4 f_{\Psi_j}(\psi(x_r, t) | \Pi_j) P(\Pi_j) \quad (7)$$

$$F_{\Psi}(\psi(x_r, t)) = \int_0^{\psi} f_{\Psi}(\psi(x_r, t)) d\psi \quad (8)$$

where $f_{\Psi}(\psi(x_r, t))$ is the total sea-level rise PDF, and $P(\Pi_j)$ is the occurrence probability of the climate change emission scenario Π_j . According to assumption [B] defined in Section 2, $P(\Pi_j)$ is 1/4. The detailed procedures to calculate the PDF of each sea-level rise component and perform the regional sea-level rise hazard assessment are shown in Fig. 2 and discussed in the next subsection.

3.1. Sterodynamic sea-level rise assessment

Sterodynamic sea-level rise is the combination of thermosteric sea-level rise due to ocean volumetric thermal expansion and ocean dynamic sea-level rise due to ocean current circulations [75,76]. The sea-level rise caused by these subcomponents can be obtained from the GCM output variables. In addition, each GCM output variable is dependent on the climate change emission scenario Π_j . In this paper, the uncertainties associated with sterodynamic sea-level rise (i.e., the sterodynamic sea-level rise PDF) are considered using 19 GCM output variables of the thermosteric and dynamic sea-level rise obtained from CMIP5 [70] given climate change emission scenario Π_j . The sterodynamic sea-level rise PDF is assumed to have a normal distribution and is validated using the Kolmogorov–Smirnov normality test (K–S test). The details of the sterodynamic sea-level rise PDF assessment are shown in Fig. 2.

Before utilizing the GCM output variables of the thermosteric and dynamic sea-level rise for estimating the sterodynamic sea-level rise

PDF, the model drift correction due to the dynamic imbalance of the GCM initial conditions is applied based on the preindustrial control experiments of the corresponding GCM through linear regression, as suggested by Jackson & Jevrejeva [27]. The inverse barometer (IB) effect induced by atmospheric pressure variations is neglected due to its insignificant contribution relative to the dynamic sea-level rise [75].

The dynamic sea level data are provided as a spatial ocean grid with different resolutions between GCMs, while the thermosteric sea-level rise data are provided as a global mean sea-level rise (i.e., the average value of all ocean grid points). Before calculating the sterodynamic sea-level rise, the dynamic sea-level data are interpolated into a latitude-longitude grid of $0.5^{\circ} \times 0.5^{\circ}$ for all models. After the ocean grid refinement and model drift correction process, the sterodynamic sea-level rise S can be calculated as [27,75,77]:

$$S_{mj}(x_r, t) = \Delta DSL_{mj}(x_r, t) + STR_{mj}(t) \quad (9)$$

$$\Delta DSL_{mj}(x_r, t) = DSL_{mj}(x_r, t) - DSL_{mj}(x_r, t_0) \quad (10)$$

where subscript m_j represents the GCM m_j ($m = 1, 2, \dots, 19$) given Π_j ($j = 1, 2, 3, 4$), $\Delta DSL_{mj}(x_r, t)$ is the dynamic sea-level change, $STR_{mj}(t)$ is the global mean thermosteric sea-level rise, and $DSL_{mj}(x_r, t)$ and $DSL_{mj}(x_r, t_0)$ are the dynamic sea levels evaluated at time t and present time t_0 in years, respectively. The global mean sterodynamic sea-level rise S equals the global mean thermosteric sea-level rise STR since the global mean dynamic sea-level rise is zero over time [75].

In Eq. (9), the sterodynamic sea-level rise does not account for the spatial variability associated with the ocean mass redistribution due to the volume expansion. This condition is known as the self-attraction and loading (SAL) effect [78] and is estimated based on ocean bottom pressure (OBP) change data using the Norwegian Earth System Model (i.

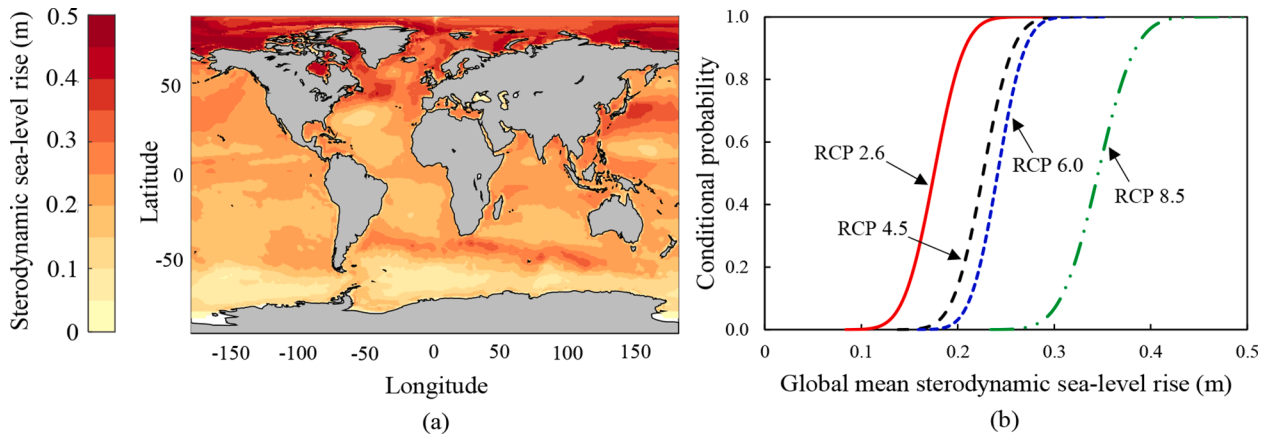


Fig. 3. Sterodynamic sea-level rise estimation results for 2100: (a) regional sterodynamic mean sea-level rise considering all RCP scenarios and (b) conditional probability of global mean sterodynamic sea-level rise given RCP scenarios (i.e., Π_j).

e., NorESM1-M) for each RCP scenario, as suggested by Richter et al. [79]. The OBP change is assumed to be a static load, and the sea-level equation solver provided by Adhikari et al. [80] is employed to translate static loading into a regional sea-level rise due to the SAL effect. The estimated regional sea-level rise induced by the SAL effect is normalized with its associated global mean sea-level rise to obtain the sterodynamic sea-level spatial variability (i.e., sea-level fingerprint) [66].

Finally, the regional sterodynamic sea-level rise ω can be calculated by scaling the sterodynamic sea-level rise S with its corresponding sea-level fingerprint FG_{STR} conditioned upon GCM m and RCP scenario j according to Eq. (3). After estimating the sterodynamic sea-level rise for all GCMs and RCP scenarios, as shown in Fig. 2, the global mean and regional sterodynamic sea-level rise PDFs are calculated through a conventional statistical analysis. Since the global mean and regional sterodynamic sea-level rise are initially assumed to be normally distributed, the K-S test is performed to validate this assumption for each estimated ocean grid cell, as suggested by Jackson & Jevrejeva [27]. During the K-S test assessment, the data distribution of all GCMs for RCP scenario j cannot be assumed to be normally distributed when the K-S test p-value is less than the significance level (i.e., 5%). Under this condition, the generalized extreme Studentized deviate (GESD) test is performed to remove the associated data outliers. After performing both tests and reverifying the data distributions, the global mean and regional sterodynamic sea-level rise PDFs can be modeled as normal distributions.

Figure 3 shows the global mean and associated regional sterodynamic sea-level rise hazard considering all RCP scenarios for 2100

relative to the present. As shown in Fig. 3a, a high sterodynamic sea-level rise is observed near the northern coast of Japan and across the Arctic Ocean. In contrast, the Southern Ocean near Antarctica manifests a smaller sea-level rise than other regions. In a moderate climate change scenario (i.e., RCP 4.5 and 6.0), the global mean sterodynamic sea-level rise is likely to exceed 0.2 m and might reach more than 0.3 m in an extreme emission scenario such as RCP 8.5.

3.2. Glacier sea-level rise assessment

Glacier sea-level rise occurs due to changes in the surface mass balance of glaciers on land and in mountain ranges owing to the seasonal snow precipitation and temperature-induced melting of ice [81,82]. In this study, the probabilistic projections of the global mean glacier sea-level rise G are obtained from Marzeion et al. [71] based on the 11 glacier models involved in the GlacierMIP.

The regional glacier sea-level rise ξ can be calculated by multiplying the global mean glacier sea-level rise G with its corresponding sea-level fingerprint FG_{GLA} according to Eq. (4). In this study, the glacier sea-level fingerprint is generated based on the land-water mass change dataset obtained from the Gravity and Recovery Climate Experiment (GRACE) [83], as suggested by Thomas & Lin [66]. The land-water mass change data used for estimating the glacier sea-level fingerprint are selected according to the glacier location identified by the Randolph Glacier Inventory [84]. After identifying the glacier location and the corresponding water mass change, the glacier sea-level fingerprint is calculated using the sea-level equation solver provided by Adhikari et al. [80]

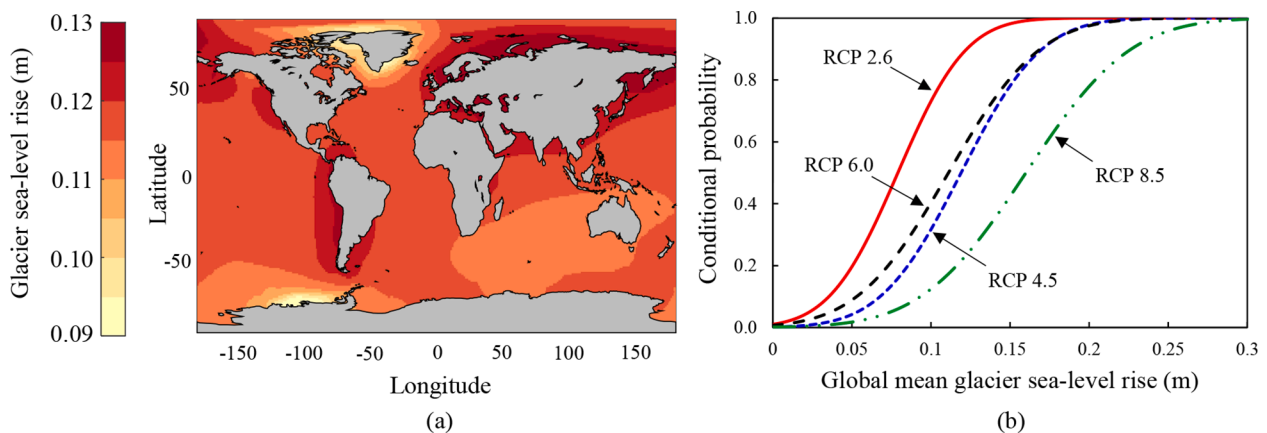


Fig. 4. Glacier sea-level rise estimation results for 2100: (a) regional glacier mean sea-level rise considering all RCP scenarios and (b) conditional probability of global mean glacier sea-level rise given RCP scenarios (i.e., Π_j).

to convert the land-water mass change into the regional sea-level change with a resolution of 0.5° for the latitude and longitude. The regional sea-level change is normalized with its associated global mean to obtain the glacier sea-level fingerprint. The regional glacier sea-level rise PDF is calculated based on MCSs, in which the probabilistic projections of global mean glacier sea-level rise G and monthly sea-level fingerprints FG_{GLA} are considered random variables, according to Thomas & Lin [66]. The land-water mass changes from the Antarctic and Greenland ice sheets are excluded from the data during glacier sea-level fingerprint estimation [84]. The surface mass balance and ice sheet dynamics of the sea-level rise are assessed separately in the ice sheet sea-level rise assessment.

Figure 4 shows the global mean glacier sea-level rise conditional probability given the RCP scenario Π_j ($j = 1, 2, 3, 4$), including the regional glacier mean sea-level rise for 2100 relative to the present. As shown in Fig. 4b, the glacier component is likely to contribute between 0.05 and 0.20 m to the global mean sea-level rise. The glacier sea-level rise for RCP 4.5 is slightly larger than that for RCP 6.0 because of the larger temperature change at the glacier location given RCP 4.5 [65,71]. In terms of the regional glacier sea-level rise, the results suggest that the sea-level rise intensity near the glaciers (e.g., Alaska, Western Canada, Southern Andes, and Russian Arctic) is high except for near the Greenland periphery and a small part of Antarctica, where most of the land-water mass decreases due to glacier melting. This phenomenon occurs because the regional glacier sea-level rise at any particular location depends on not only the local land-water mass change but also the gravitational influence between the glacier ice melts and surrounding ocean [81].

3.3. Ice sheet sea-level rise assessment

The ice sheet sea-level rise can be attributed to the two largest ice masses: the Greenland ice sheet (GrIS) and Antarctic ice sheet (AIS). Each ice sheet contributes to the sea-level rise through the surface mass balance and dynamic change component associated with subshelf melting processes [25]. In this paper, the ice sheet dynamics and surface mass balance are evaluated through sensitivity experiments based on six ice sheet models (ISMs) involved in the SeaRISE project, as summarized by Bindshadler et al. [72]. Before calculating the ice sheet sea-level rise, several types of data associated with ocean surface temperature and near-surface atmospheric temperature are obtained from the GCM output variables. In addition, a response-function-based method developed by Levermann et al. [85] is used to evaluate the ice sheet dynamic sea-level rise, while the surface mass balance sea-level rise is estimated based on the volume above floatation (VAF) change data [72]. In the ice sheet sea-level rise assessment, MCSs are performed, as shown in Fig. 2. The associated random variables are listed in Table 1.

The ice sheet dynamic sea-level rise for the AIS and GrIS is calculated by converting the global mean ocean surface temperature obtained from the associated GCM output variables to a subsurface ocean temperature. The subsurface ocean temperature is assessed differently for the GrIS and AIS. In terms of the AIS subsurface ocean temperature, a complete scaling coefficient for every GCM and four basins in the AIS was provided by Levermann et al. [85] as:

$$\Delta\tau_{A,mj}(t) = \alpha\Delta\tau_{mj}(t) \quad (11)$$

where $\Delta\tau_{A,mj}$ is the AIS subsurface ocean warming at time t in years with respect to the present time conditioned upon GCM m and Π_j (i.e., m_j), α is the temperature scaling coefficient, and $\Delta\tau_{mj}$ is the global mean ocean surface temperature warming given GCM m and Π_j .

Due to data scarcity, the subsurface ocean warming data for the GrIS are estimated directly from the subsurface ocean warming projections under scenario A1B of the Special Report on Emissions Scenarios (SRES) provided by Yin et al. [86]. The climate change scenario of the SRES A1B is similar to the RCP 6.0 scenario, according to previous studies [25,87].

In addition, the ice sheet mass loss can be considered to be linearly related to the temperature rise, according to Le Bars et al. [74]. Therefore, to consider the uncertainties associated with the global mean ocean surface temperature given a GCM and an RCP scenario, the subsurface ocean temperature defined by Yin et al. [86] is modified as:

$$SF_{mj}(t) = \frac{\Delta\tau_{mj}(t)}{\Delta\tau_{m(RCP6.0)}(t)} \quad (12)$$

$$\Delta\tau_{G,mj}(t) = SF_{mj}(t) \cdot \Delta\tau_{Gref}(t) \quad (13)$$

where SF_{mj} is the scale factor associated with the global mean ocean surface temperature from GCM m given RCP scenario j , $\Delta\tau_{m(RCP6.0)}$ is the global mean ocean surface temperature warming under the RCP 6.0 scenario from the corresponding GCM m , $\Delta\tau_{G,mj}$ is the subsurface ocean warming for Greenland given GCM m and Π_j , and $\Delta\tau_{Gref}$ is the subsurface ocean warming presented by Yin et al. [86].

The obtained subsurface ocean warming is utilized to evaluate the basal melting for each ice sheet by using an empirical basal melt coefficient [85] expressed as:

$$\Delta b_{A,mj}(t) = \beta\Delta\tau_{A,mj}(t) \quad (14)$$

$$\Delta b_{G,mj}(t) = \beta\Delta\tau_{G,mj}(t) \quad (15)$$

where $\Delta b_{A,mj}$ and $\Delta b_{G,mj}$ are the basal melting for the AIS and GrIS given GCM m and Π_j , respectively, and β is the basal melt sensitivity selected randomly from a uniform interval of $7 \text{ m yr}^{-1}\text{K}^{-1}$ to $16 \text{ m yr}^{-1}\text{K}^{-1}$ [85].

The response function representing the ratio between the subshelf ice discharge rate and basal melt rate can be derived based on the sensitivity parameter obtained by using the equation proposed by Levermann et al. [85] as:

$$R_{A,l,bs}(t) = \frac{1}{\Delta m_0} \frac{dB_{A,l,bs}(t)}{dt} \quad (16)$$

$$R_{G,l,bs}(t) = \frac{1}{\Delta m_0} \frac{dB_{G,l,bs}(t)}{dt} \quad (17)$$

where subscripts l and bs are the ISM number and analyzed basin, respectively, $R_{A,l,bs}(t)$ and $R_{G,l,bs}(t)$ are the response functions for the AIS and GrIS given ISM l ($l = 1, 2, \dots, 6$), respectively, $B_{A,l,bs}(t)$ and $B_{G,l,bs}(t)$ are the subshelf ice discharges in the AIS and GrIS given ISM l , respectively, and Δm_0 is the basal melt rate with a value of 20 m yr^{-1} according to the SeaRISE M2 experiment [72]. In addition, the subshelf ice discharge is obtained from the SeaRISE sensitivity experiments and converted from volume above floatation to the global mean sea-level equivalent [72,88,89].

Finally, the ice sheet dynamic sea-level rise from all basin locations is calculated by integrating the ice sheet basal melting with its corresponding response function over time as:

$$I_{A,dyn,mj,l}(t) = \sum_{bs=1}^{nba} \int_0^t R_{A,l,bs}(t) \Delta b_{A,mj}(t) dt \quad (18)$$

$$I_{G,dyn,mj,l}(t) = \sum_{bs=1}^{nbg} \int_0^t R_{G,l,bs}(t) \Delta b_{G,mj}(t) dt \quad (19)$$

where $I_{A,dyn,mj,l}$ and $I_{G,dyn,mj,l}$ are the ice sheet dynamic sea-level rises from the AIS and GrIS, given GCM m , Π_j , and ISM l , respectively, and nba and nbg are the total number of analyzed basins in the AIS and GrIS, respectively. In Eqs. (18) and (19), the integration over time is performed relative to the present.

The surface mass balance component of the ice sheet sea-level rise is calculated based on the VAF change data from the results of the SeaRISE C1 experiment and converted into a global mean sea-level rise equivalent [72]. The uncertainties associated with the VAF change are

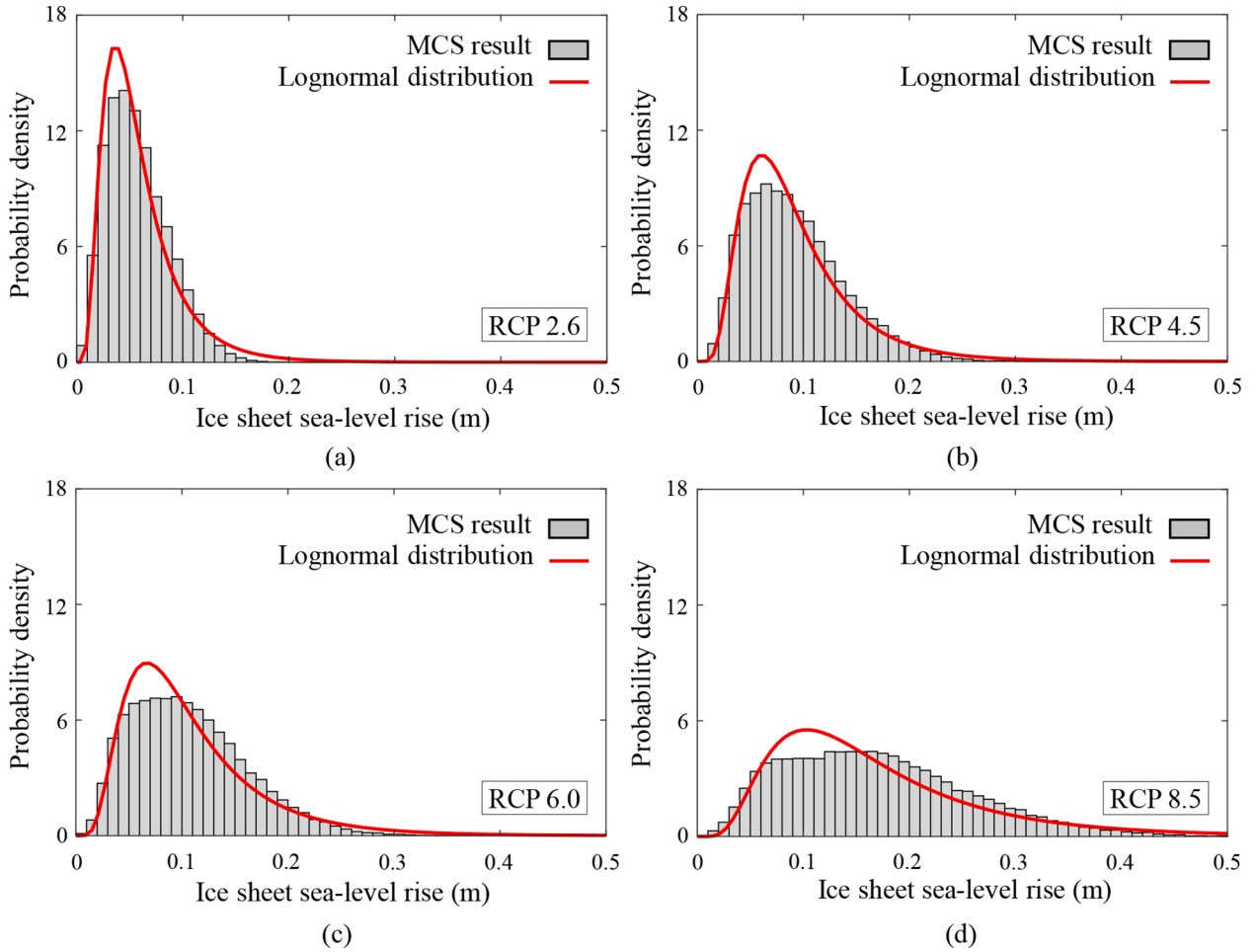


Fig. 5. MCS results and conditional PDFs associated with global mean ice sheet sea-level rises given RCP scenarios (i.e., Π_j): (a) RCP 2.6, (b) RCP 4.5, (c) RCP 6.0, and (d) RCP 8.5.

considered using the six ISMs involved in the SeaRISE C1 experiment. In addition, the SeaRISE C1 experiment was conducted based on the global mean surface temperature warming under the climate change scenario of SRES A1B, which is similar to the RCP 6.0 scenario, according to previous studies [25,87]. To consider the uncertainties associated with the global mean surface temperature conditioned upon a GCM and an RCP scenario, the VAF changes for the AIS and GrIS are modified based on a scale factor in a similar manner as the subsurface ocean temperatures as:

$$SF_{a,mj}(t) = \frac{\Delta\tau_{a,mj}(t)}{\Delta\tau_{a,m(RCP6.0)}(t)} \quad (20)$$

$$I_{A_{smb},mj,l}(t) = SF_{a,mj}(t) \cdot \frac{\Delta VAF_{A,l}(t)}{4 \times 10^{14}} \quad (21)$$

$$I_{G_{smb},mj,l}(t) = SF_{a,mj}(t) \cdot \frac{\Delta VAF_{G,l}(t)}{4 \times 10^{14}} \quad (22)$$

where $SF_{a,mj}$ is the scale factor associated with the global mean surface temperature conditioned upon GCM m and RCP scenario j , $\Delta\tau_{a,mj}$ is the global mean surface temperature warming, $\Delta\tau_{a,m(RCP6.0)}$ is the global mean surface temperature warming given the RCP 6.0 scenario and the corresponding GCM m , $I_{A_{smb},mj,l}$ and $I_{G_{smb},mj,l}$ are the surface mass balance ice sheet sea-level rises from the AIS and GrIS given GCM m , Π_j , and ISM l , respectively, and $\Delta VAF_{A,l}$ and $\Delta VAF_{G,l}$ are the volume changes above floatation in m^3 from the AIS and GrIS given ISM l , respectively.

The final global mean ice sheet sea-level rise is a simple summation

of the dynamic and surface mass balance components from the AIS and GrIS as:

$$I_{mj,l}(t) = I_{A_{dyn},mj,l}(t) + I_{G_{dyn},mj,l}(t) + I_{A_{smb},mj,l}(t) + I_{G_{smb},mj,l}(t) \quad (23)$$

where $I_{mj,l}$ is the global mean ice sheet sea-level rise given GCM m , Π_j , and ISM l .

An MCS is performed to estimate the PDF of the global mean ice sheet sea-level rise considering the uncertainties associated with the global mean ocean temperature, surface temperature, basal melt sensitivity, RCP scenarios, and ISM, as shown in Fig. 2. The regional ice sheet sea-level rise η can be calculated by multiplying the global mean ice sheet sea-level rise I with its corresponding sea-level fingerprint FG_{ICS} according to Eq. (5). The ice sheet sea-level fingerprint is generated in a manner similar to the glacier sea-level fingerprint, by using the land-water mass change dataset from the GRACE experiment [83]. However, the land-water mass change data used for estimating ice sheet sea-level fingerprints are selected only for the Greenland and Antarctic ice sheets. As shown in Fig. 2, the MCS is performed according to Eq. (5) to assess the PDF associated with the regional ice sheet sea-level rise considering the uncertainty associated with the sea-level fingerprint and global mean ice sheet sea-level rise.

Figure 5 shows the MCS results of the global mean ice sheet sea-level rise for all RCP scenarios in 2100 relative to the present. Based on the MCS results, the ice sheet sea-level rise PDF is approximated as a lognormal distribution. As shown in Fig. 5, the average value varies considerably from 0.07 to 0.18 m, with a broader data spread as the RCP emissions increase. This phenomenon occurs because the variability in

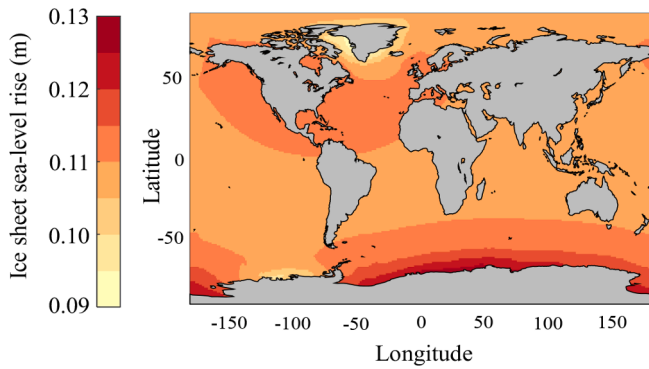


Fig. 6. Regional ice sheet mean sea-level rise considering all RCP scenarios for 2100.

the temperature data associated with the considered GCMs increases among models as the RCP emission increases. Computational results of the regional ice sheet mean sea-level rise indicate a relatively constant sea-level rise across the ocean with a value between 0.1 and 0.12 m, as shown in Fig. 6. In addition, the melting rate in the West Antarctica region, particularly in the Amundsen Sea, is greater than that in the other regions, while eastern Antarctica indicates a positive trend due to the intensified climate-induced snow precipitation [90].

3.4. Total global mean and regional sea-level rise hazard

Figure 7 shows the conditional probability of total global mean sea-level rise for all components in 2100 relative to 2020 given the RCP (i.e., Π_j) and the time-series projection of each sea-level rise component. The total global mean sea-level rise CDF displayed in Fig. 7a is calculated by summing the CDFs from the global mean sea-level rise components estimated using the MCS. Considering assumption [B], the global mean total sea-level rise hazard is between 0.4 and 0.5 m. Moreover, the comparison of Figs. 3, 4, and 6 indicates that the sterodynamic sea-level rise due to the volumetric thermal expansion is the primary driving force of the overall sea-level rise compared with ice melting. Conversely, the glacier and ice sheet sea-level rise suggest a lower global impact with a more uniform sea-level rise over the ocean. In terms of the time-series projection, as shown in Fig. 7b, the total sea-level rise, including its components, indicates a relatively linear trend over time, which verifies the previous assumption that the sea-level rise components are perfectly correlated with one another.

Although probabilistic projections of the sea-level rise have been performed by many studies [e.g., [24–27,69,73]], the variability in the regional sea-level rise for the four RCP scenarios remains insufficiently provided. In addition, several of these studies rely heavily on the experts' elicitation for estimating the uncertainties associated with the ice sheet sea-level rise and assume mutually exclusive processes between the sea-level rise components [e.g., [26,27,69]]. This paper aims to reconstruct the probabilistic sea-level rise assessment based on a more holistic approach by combining the sea-level rise predictions from the

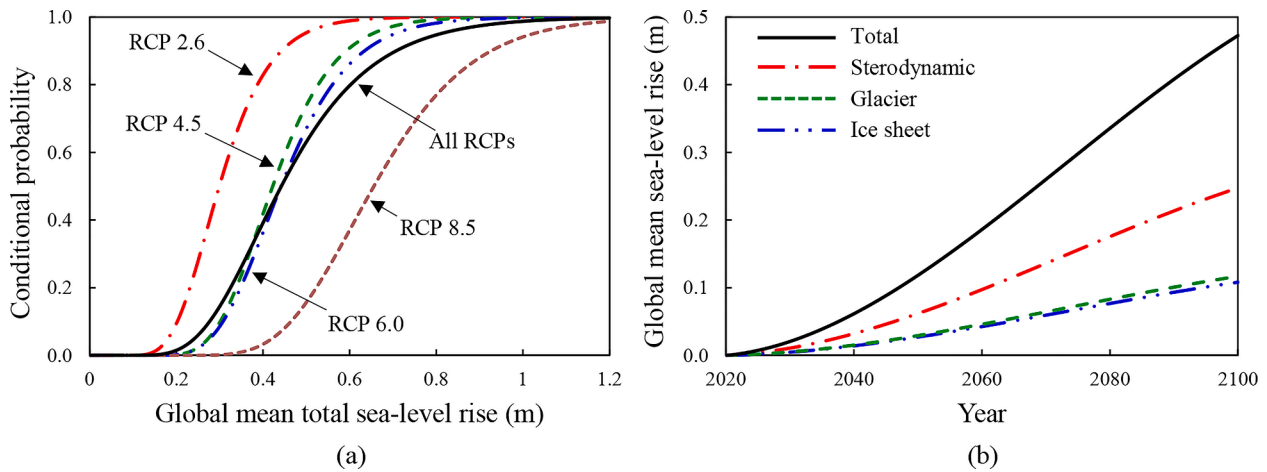


Fig. 7. (a) Conditional probability of total global mean sea-level rise in 2100 given RCP scenarios Π_j and (b) time-series projection of global mean sea-level rise for each component considering all RCP scenarios.

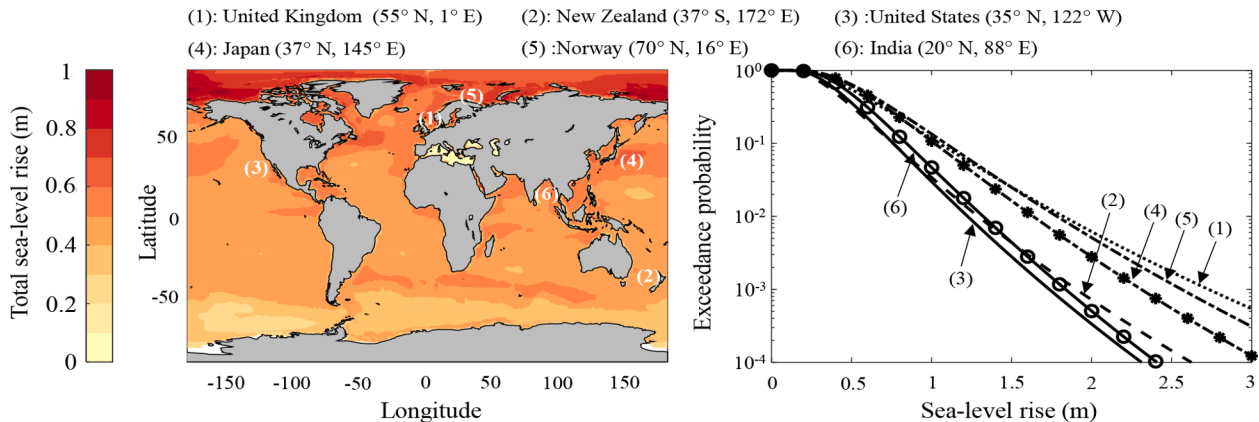


Fig. 8. Example of regional sea-level rise hazard projected in 2100: (1) United Kingdom, (2) New Zealand, (3) United States, (4) Japan, (5) Norway, and (6) India.

available models and assuming a perfect correlation between the sea-level rise components to comprehend the variability associated with the regional sea-level rise. The variability in the regional sea-level rise is a crucial aspect for assessing the regional sea-level rise hazard.

Figure 8 shows the total regional sea-level rise hazard and the associated curves under assumption [B] for several countries located near different oceans. Countries and areas situated near the Indian Ocean and Pacific Ocean, such as India, New Zealand, and the west coast of the United States, indicate lower sea-level rise hazards. However, an exception is observed for the northern coast of Japan, where the sea-level rise hazard is the highest among areas located in the same ocean. In the case of high-latitude areas located near the Arctic Ocean and North Atlantic Ocean, such as the west coast of the United Kingdom (i.e., the North Sea) and northeastern coast of Norway, the sea-level rise hazard is considerably higher than that of the regions located near other oceans. The high-resolution hazard map associated with the sea-level rise is a prognostic tool for determining the hazard intensity over an earthquake fault. To the best of the authors' knowledge, none of the existing studies have examined the sea-level rise hazard while considering the spatial variability and climate change severity in the tsunami hazard assessment. The sea-level rise hazard map provided in this paper can be used as a reference in investigating the aggravation in tsunami hazards due to sea-level rise in different regions worldwide.

4. Probabilistic tsunami hazard assessment

To clarify the relationship between the exceedance probability and tsunami inundation depth (i.e., tsunami hazard curve), MCS-based horizontal 2D tsunami propagation simulations based on the nonlinear longwave theory have been performed considering the uncertainty associated with earthquake fault parameters (i.e., the average stress drop and rake angle) [16,17]. However, horizontal 2D tsunami propagation simulations incur a large computational cost. In addition, since the conditional tsunami hazards are estimated for the various sea-level rises shown in Fig. 1, an enormous number of MCS samples is required. Therefore, a procedure for conditional tsunami hazard assessment given sea-level rise is developed herein based on the RBF network and quasi-MCS for reducing the associated numerical simulation costs.

4.1. Tsunami propagation analysis considering sea-level rise

The initial tsunami height for the tsunami propagation analysis is computed by employing the formula associated with the slip dislocation and rake angle developed by Okada [91]. Next, a horizontal 2D tsunami propagation simulation is performed based on nonlinear longwave theory to spatially evaluate the inundation depth for 30 m grid cells [92]. The sea-level rise effect on the tsunami propagation analysis is taken into account by decreasing the bathymetric depth and topographic elevation before the simulation is performed. The land is considered inundated due to sea-level rise when the corresponding elevation is negative (i.e., less than the mean sea level). The corresponding roughness coefficient for the inundated land area is modified similar to that of the ocean roughness coefficient [93]. After the propagation analysis, the tsunami inundation depth is determined by subtracting the tsunami maximum wave height measured from the mean sea level from the land elevation. Finally, the tsunami hazard assessment conditioned upon sea-level rise is performed considering the uncertainties associated with the fault movement parameters.

4.2. Radial basis function network surrogate model

The RBF network has been commonly used for the adaptive approximation of a function and has been mainly applied to data with nonlinear characteristics [94,95]. The RBF network covers three layers of a network: an input layer with q nodes, a hidden layer with k RBFs,

and an output layer with one node [96]. In this study, a widely used Gaussian RBF transfer function is employed for estimating the RBF network response surface. The three-layered structure of an RBF network associated with the inundation depth can be quantitatively expressed as follows:

$$g_p(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_p\|^2}{r^2}\right) \quad (24)$$

$$h(\mathbf{x}) = \sum_{p=1}^k \lambda_p \cdot g_p(\mathbf{x}) \quad (25)$$

where $g_p(\mathbf{x})$ is the output data from the p -th RBF in the hidden layer (i.e., the Gaussian RBF transfer function). Here, $\mathbf{x} (=x_1, \dots, x_q)$ represents the input data vector from the input elements linked to the p -th RBF in the hidden layer, r is the associated shape parameter of the RBF, $\mathbf{x}_p$ is the center point of the RBF, which is set for each data point, $h(\mathbf{x})$ is the value of the output layer node (i.e., inundation depth), and λ_p is the weighting factor of the p -th RBF with a total of k sets of RBFs.

By utilizing k sets of training data $\mathbf{x}_p$ and supervised data y_p ($p = 1, 2, \dots, k$), the weighting factor λ ($=\lambda_1, \lambda_2, \dots, \lambda_k$) can be derived by solving the linear system as:

$$\lambda = \mathbf{G}^{-1} \mathbf{y} \quad (26)$$

where:

$$\mathbf{G} = \begin{pmatrix} g_1(\mathbf{x}_1) & g_2(\mathbf{x}_1) & \dots & g_k(\mathbf{x}_1) \\ g_1(\mathbf{x}_2) & g_2(\mathbf{x}_2) & \dots & g_k(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ g_1(\mathbf{x}_k) & g_2(\mathbf{x}_k) & \dots & g_k(\mathbf{x}_k) \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_k \end{pmatrix} \quad (27)$$

The optimal shape parameter of an RBF network can be obtained using the leave-one-out cross-validation (LOOCV) technique proposed by Rippe [97]. The underlying principle of the LOOCV is to remove one training dataset $\mathbf{x}_i$ to obtain an incomplete training dataset of $\mathbf{x}_{p-i}$ ($p-i = 1, 2, \dots, i-1, i+1, \dots, k$) and construct a new RBF response surface by using the incomplete training dataset for model verification. The root mean square error for the LOOCV is:

$$RMSE = \sqrt{\frac{1}{k} \sum_{p=1}^k (y_p - h_{[-i]}(\mathbf{x}_p))^2} \quad (28)$$

where $h_{[-i]}(\mathbf{x}_p)$ is the inundation depth estimated based on the RBF network response surface obtained using the incomplete training dataset of $\mathbf{x}_{p-i}$ ($p-i = 1, 2, \dots, i-1, i+1, \dots, k$).

Hyperparameter optimization of shape parameter r is performed using a grid search algorithm to obtain the optimum shape parameter by minimizing the averaged RMSE value from the LOOCV of all training data [98].

4.3. Conditional tsunami hazard assessment given sea-level rise

Conditional tsunami hazard curves are developed based on the RBF network response surface and fault parameter uncertainties by utilizing a quasi-Monte Carlo method due to its effectiveness and rapid convergence for numerical integration [99]. The Faure sequence is applied in the MCS as a deterministic alternative to generate random numbers [63]. Based on the uncertainties associated with the prediction of fault movement, a random combination of fault parameters is generated by performing inverse transform sampling [100]. The tsunami inundation depth is evaluated from the corresponding fault parameters by using the associated RBF network response surface.

Finally, the inundation depth is rearranged according to its frequency of occurrence to form a conditional tsunami hazard curve given a sea-level rise. In this paper, the surrogate model used for tsunami hazard assessment is validated through a conventional MCS, as

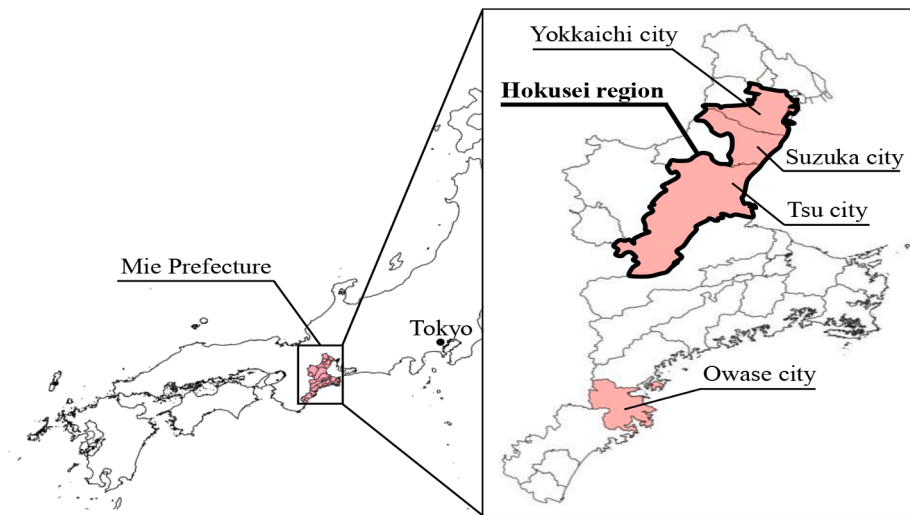


Fig. 9. Analyzed municipalities in Mie Prefecture, Japan.

discussed in the following case study.

5. Illustrative example

5.1. Anticipated Nankai-Tonankai earthquake

The proposed framework is applied for several municipalities in the Mie Prefecture of Japan, namely, the Hokusei region, including Yokkaichi city, Suzuka city, Tsu city, and Owase city, as shown in Fig. 9. In this case study, the impacts of sea-level rise on tsunamis generated by

the anticipated Nankai-Tonankai earthquake are investigated. Eleven tsunami fault movement models for the Nankai-Tonankai earthquake have been provided by the Cabinet Office, Government of Japan [101], predicted based on historical earthquakes with moment magnitudes higher than 8.0 (i.e., the 1944 Tonankai earthquake, 1946 Nankai earthquake, 2003 Tokachi-Oki earthquake, 2004 Sumatra-Andaman earthquake, 2010 Maule earthquake, and 2011 Great East Japan earthquake). In this paper, the fault model shown in Fig. 10 is used for the tsunami hazard assessment. Note that the fault rupture areas are determined following the assumption of an average stress drop made by Ishibashi et al. [16,18]. In addition, supplemental data such as bathymetry and roughness coefficients are obtained by the Cabinet Office, Government of Japan [101], while the roughness coefficients are determined according to Kotani, Imamura, & Shuto [102]. Although tidal conditions affect the tsunami inundation depth [58], incorporating the stochastic nature of the tide level into the probabilistic analysis of tsunami hazards can be an intricate task since the level considerably fluctuates over time. Therefore, the tsunami analysis in this illustrative example is performed under mean sea-level conditions. As stated in assumption [E], land subsidence (i.e., coseismic deformation in the land) is excluded from the analysis to examine the sea-level rise effects on the tsunami propagation.

5.2. Probabilistic sea-level rise hazard over the anticipated Nankai-Tonankai fault movement

Figure 11 shows the sea-level rise hazard evaluated over the anticipated Nankai-Tonankai fault movement. During the stereodynamic sea-

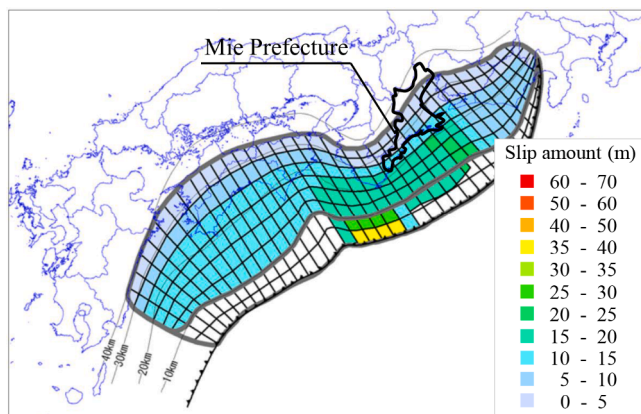


Fig. 10. Tsunami fault movement model used in this study [16,17].

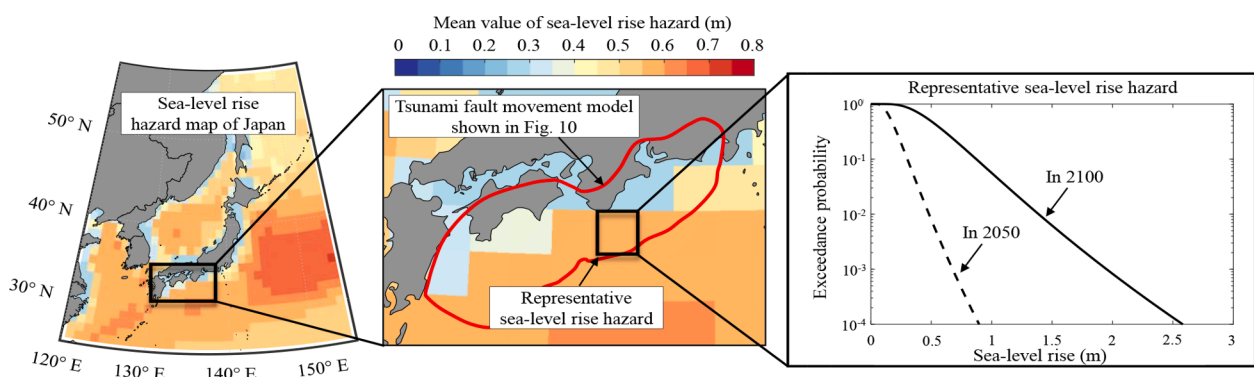


Fig. 11. Sea-level rise hazard over the anticipated Nankai-Tonankai fault movement in 2050 and 2100.

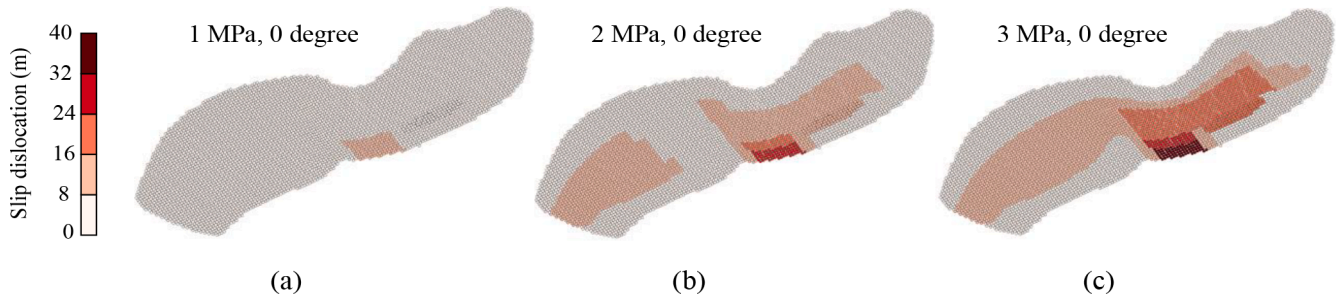


Fig. 12. Slip dislocation corresponding to the average stress drop: (a) 1 MPa, (b) 2 MPa, and (c) 3 MPa.

level rise assessment, several GCMs exhibit certain insufficiencies in simulating the dynamic sea-level rise in coastal regions due to many factors associated with the GCM process, including coarse resolution and coastal current influences [26]. Therefore, the estimated sea-level rises in the ocean grid cells near coasts are notably lower than those in the

open ocean, as outlined in Fig. 11. To resolve this issue, the sea-level rise hazard is represented by the ocean grid cell having maximum sea-level rise hazard among the ocean grid cells over the anticipated Nankai-Tonankai fault movement. The maximum sea-level rise hazard location x_r over Nankai-Tonankai pertains to 33° N and 136° E. Based on this

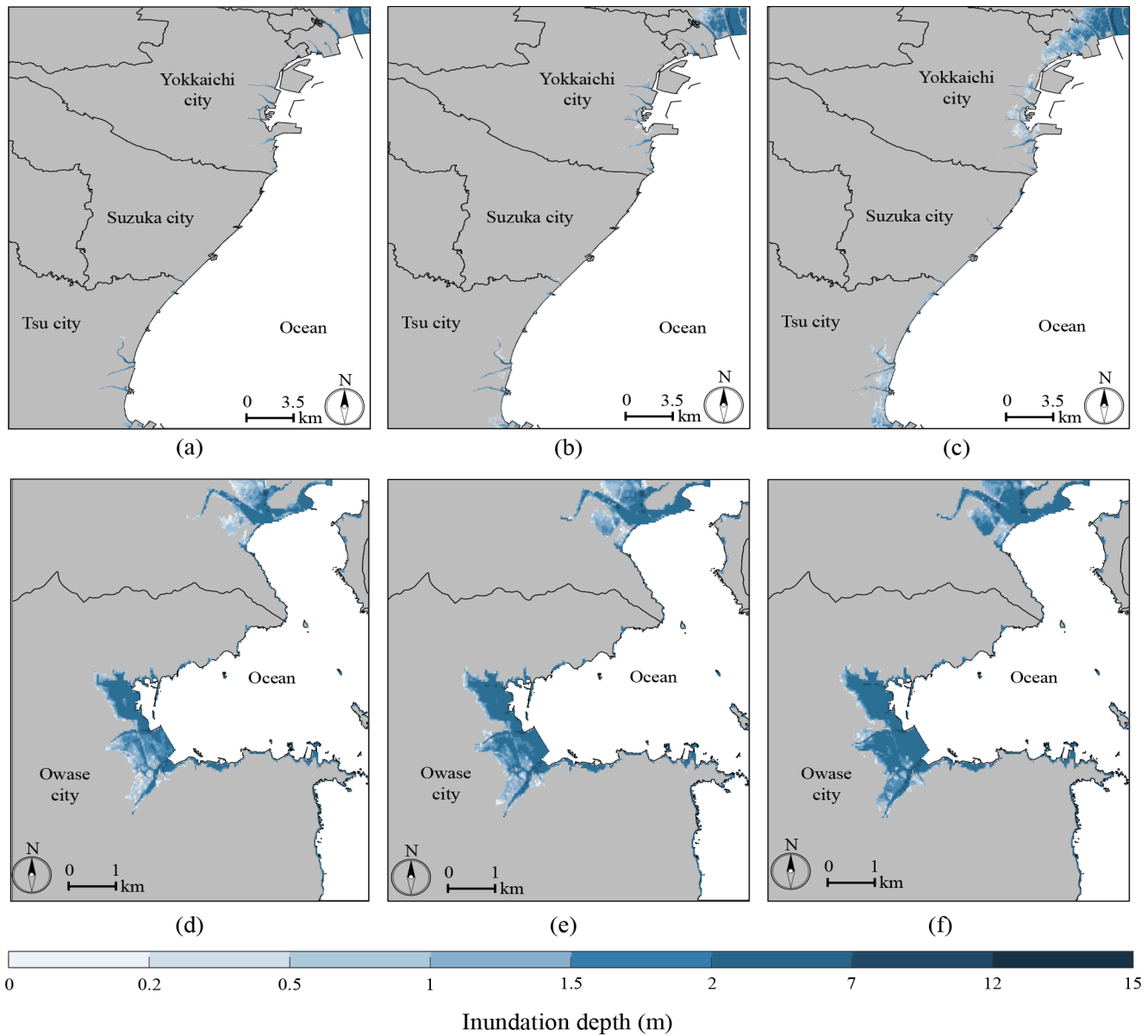


Fig. 13. Tsunami propagation analysis results for various cases associated with sea-level rise under a stress drop of 2 MPa and additional rake angle of 10° : (a) Hokusei region with current sea level, (b) Hokusei region with 0.6 m sea-level rise, (c) Hokusei region with 1.2 m sea-level rise, (d) Owase city with current sea level, (e) Owase city with 0.6 m sea-level rise, and (f) Owase city with 1.2 m sea-level rise.

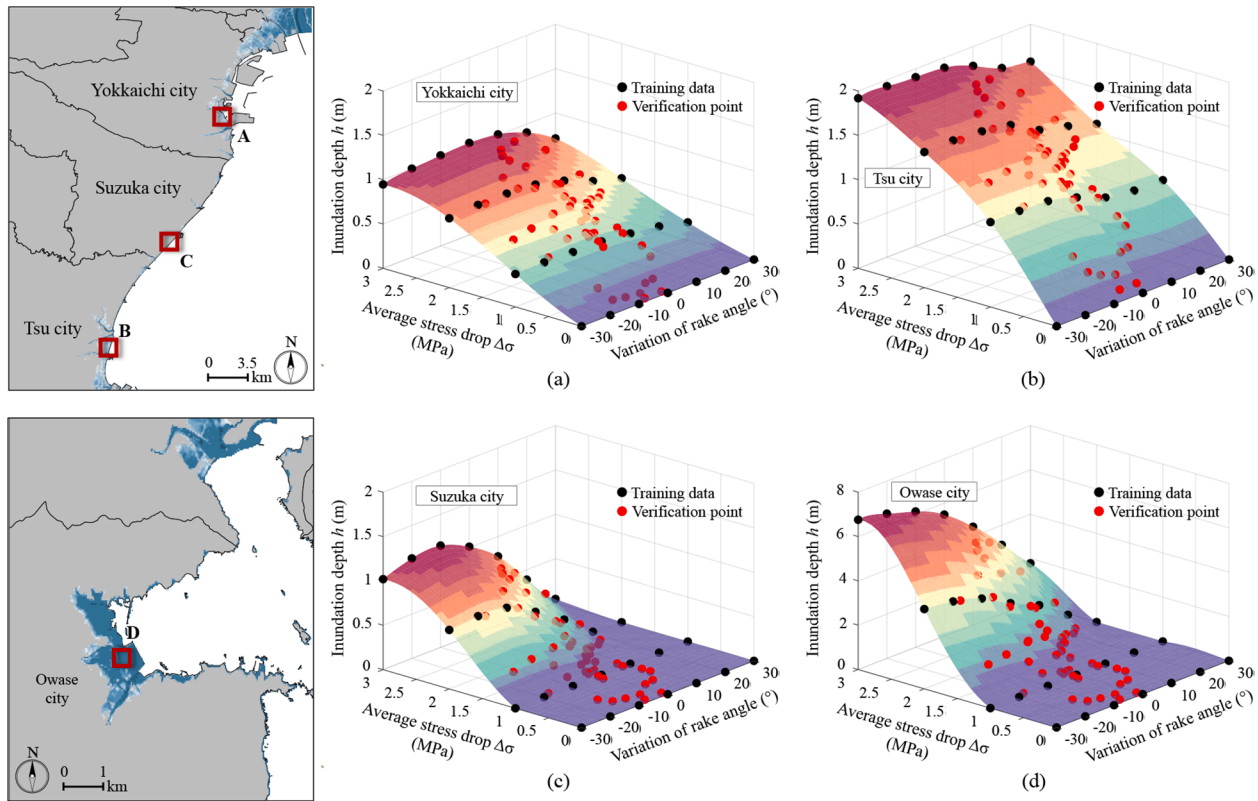


Fig. 14. RBF network response surface examples without sea-level rise in the analyzed municipalities: (a) Yokkaichi city, (b) Tsu city, (c) Suzuka city, and (d) Owase city.

assumption, as shown in Fig. 11, the mean sea-level rises over the Nankai-Tonankai in 2050 and 2100 are 0.17 m and 0.51 m, respectively.

5.3. Sea-level rise effects on tsunami hazard

The tsunami hazard assessment in Mie Prefecture is performed by identifying the uncertainties associated with the fault parameters. In this illustrative example, the average stress drop and rake angle are considered as the fault movement uncertainties. The slip dislocation is calculated from the average stress drop according to the Headquarters for Earthquake Research Promotion [103] as:

$$D_{if} = \frac{16\Delta\sigma S_f^{0.5}}{\mu 7\pi^{1.5}} \quad (29)$$

where D_{if} is the slip dislocation, S_f is the fault area, μ is the fault rigidity, and $\Delta\sigma$ is the average stress drop assumed to have a lognormal distribution with a mean of 1.2 MPa and standard deviation of 1 MPa [101]. The slip displacement in the fault model shown in Fig. 10 is estimated assuming an average stress drop of 3 MPa [101]. Since the fault movement model presented in Fig. 10 was developed under the worst-case scenario, the distribution of the average stress drop is truncated at 0 MPa and 3 MPa as lower and upper bounds, respectively, to consider the physically plausible fault movement during the tsunami hazard assessment. Moreover, it is assumed that independent of the average stress drop, the slip displacement can be estimated using Eq. (29).

In addition, the rake angle is assumed to be a random variable that follows a normal distribution with a mean value extracted from the fault model from the Cabinet Office, Government of Japan [101] and a standard deviation of 10° . According to the fault model, the rake angle value varies by approximately 90° , which indicates that the overall fault movement resembles a reverse fault type. These uncertainties are employed in the MCS for the tsunami hazard curve assessment and are discussed in the following subsection.

5.3.1. Tsunami propagation analysis in Mie Prefecture

In the tsunami inundation analysis [101], the coastal defenses are assumed to be damaged due to liquefaction prior to tsunami arrival. Following this assumption, the tsunami propagation analysis is performed without considering the coastal defenses. Before the tsunami propagation analyses are performed, the bathymetry and elevation data are decreased from 0 to 1.4 m in an interval of 0.2 m to simulate 8 different sea-level rise cases. In addition, a tsunami propagation analysis is conducted for each sea-level rise case under various fault parameter combinations. The average stress drops are sampled from 0 to 3 MPa with an interval of 1 MPa. Fig. 12 illustrates the slip dislocation associated with the fault model, assuming an average stress drop of 1 MPa, 2 MPa, and 3 MPa. As shown in Fig. 12, the slip displacement in each fault segment increases in proportion with the average stress drop based on Eq. (29). It should be noted that Fig. 12c is similar to Fig. 10.

The rake angle provided by the Cabinet Office, Government of Japan [101] is modified and sampled by adding value from -30° to 30° in an interval of 10° in the fault model based on the uncertainties associated with the additional variation in the rake angle. This modification results in 28 fault parameter combinations for each sea-level rise case considered.

Figure 13 shows the tsunami inundation analysis results obtained considering an average stress drop of 2 MPa and an additional variation of 10° in the rake angle given sea-level rises of 0, 0.6, and 1.2 m corresponding to exceedance probabilities of 1.0, 0.33, and 0.02, respectively. As shown in Fig. 13a to c, the inundation depth in the Hokusei region slightly increases, and the tsunami-affected area extends up to 1.2 km from the shore under a sea-level rise of 1.2 m. In contrast, the tsunami inundation depth in Owase city is exceptionally higher than that in the Hokusei region, as shown in Fig. 13d to f. In Owase city, a higher tsunami inundation depth tends to develop through the rivers into the city (i.e., the inner region of Owase city) as the sea level increases.

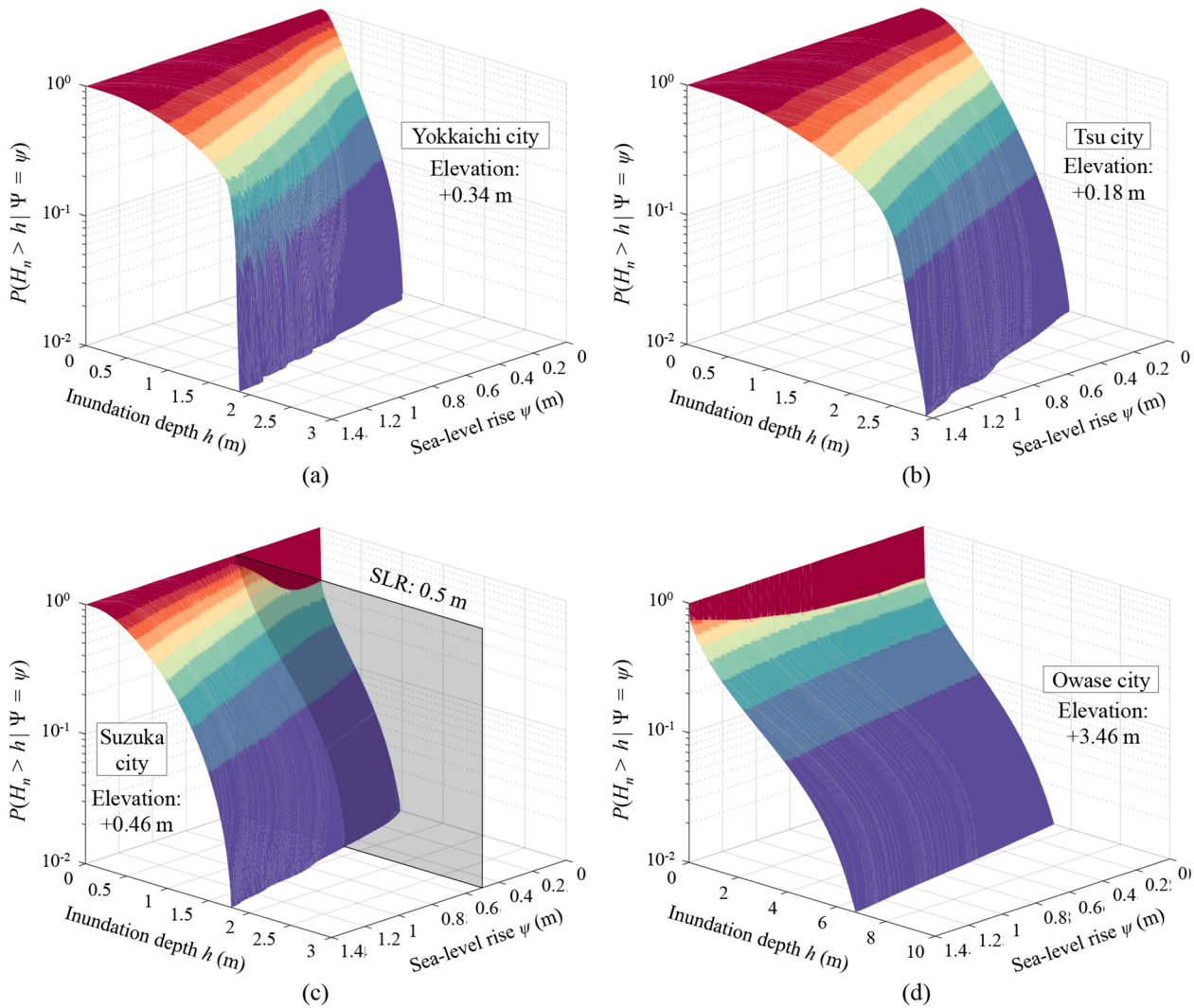


Fig. 15. Conditional tsunami hazard curves given sea-level rise in different municipalities: (a) Yokkaichi city, (b) Tsu city, (c) Suzuka city, and (d) Owase city.

The tsunami-affected area in Owase city is relatively higher than that in Hokusei region. This is because despite being situated in a topographically higher region, Owase city has a smaller distance to the earthquake fault than the Hokusei region. In addition, the nearly V-shaped coastlines in Owase city can easily accumulate and amplify the associated tsunami wave heights [104]. Regardless of the distance between the earthquake fault, the tsunami caused by the anticipated Nankai-Tonankai earthquake would generate a more extensive inundation area in the Hokusei region due to the wide plains and lower elevation under sea-level rise effects, as shown in Fig. 13c. Overall, the tsunami intensity considerably depends on the distance of the analyzed area to the earthquake fault, sea-level rise intensity, and geographical conditions of the associated coastal regions.

5.3.2. RBF network response surface assessment

Figure 14 illustrates the RBF network response surfaces estimated for a zero sea-level rise case for several locations in the Hokusei region and Owase city. The plotted black points represent the inundation depths obtained from the tsunami propagation analysis, which are employed as the training data. According to the response surface results, the inundation depth among municipalities varies according to the location and geographical conditions. Additionally, the inundation depth on the analyzed grid tends to increase as the average stress drop increases. To verify these RBF network response surfaces, 50 random inundation depth points are generated from the tsunami propagation analysis,

displayed as red points in Fig. 14. The random inundation points are compared with the response function at the same input, and the associated mean absolute error is measured. Considering the relatively small mean absolute errors of 0.04, 0.03, 0.05, and 0.17 m for Yokkaichi, Tsu, Suzuka, and Owase cities, respectively, the inundation depth can be estimated accurately using the RBF network response surfaces.

5.3.3. Tsunami hazard assessment in Mie Prefecture

For each sea-level rise case, a tsunami hazard curve is generated using a quasi-Monte-Carlo method with 100,000 simulations based on the RBF network response surface and uncertainties associated with the fault parameters (i.e., average stress drop and additional variation in the rake angle). Fig. 15 shows the conditional tsunami hazard curves with respect to sea-level rise among the four analyzed areas in the Hokusei region and Owase city, evaluated in a similar analyzed grid as that outlined in Fig. 14. The analyzed grids are situated in the land for Yokkaichi city, and in the shoreline for Suzuka and Tsu city. At a high exceedance probability (i.e., low tsunami hazard intensity), Suzuka city exhibits a significant increase in the tsunami hazard intensity when the sea-level rise exceeds the corresponding land elevation, as shown in Fig. 15c, where the sharp increase occurs at 0.5 m for Suzuka city. At a low exceedance probability, the tsunami inundation depth increases approximately linearly with the sea-level rise since the tsunami is the major contributor to the inundation depth. Furthermore, as shown in Fig. 15d, only minimal effects of the sea-level rise are observed in Owase

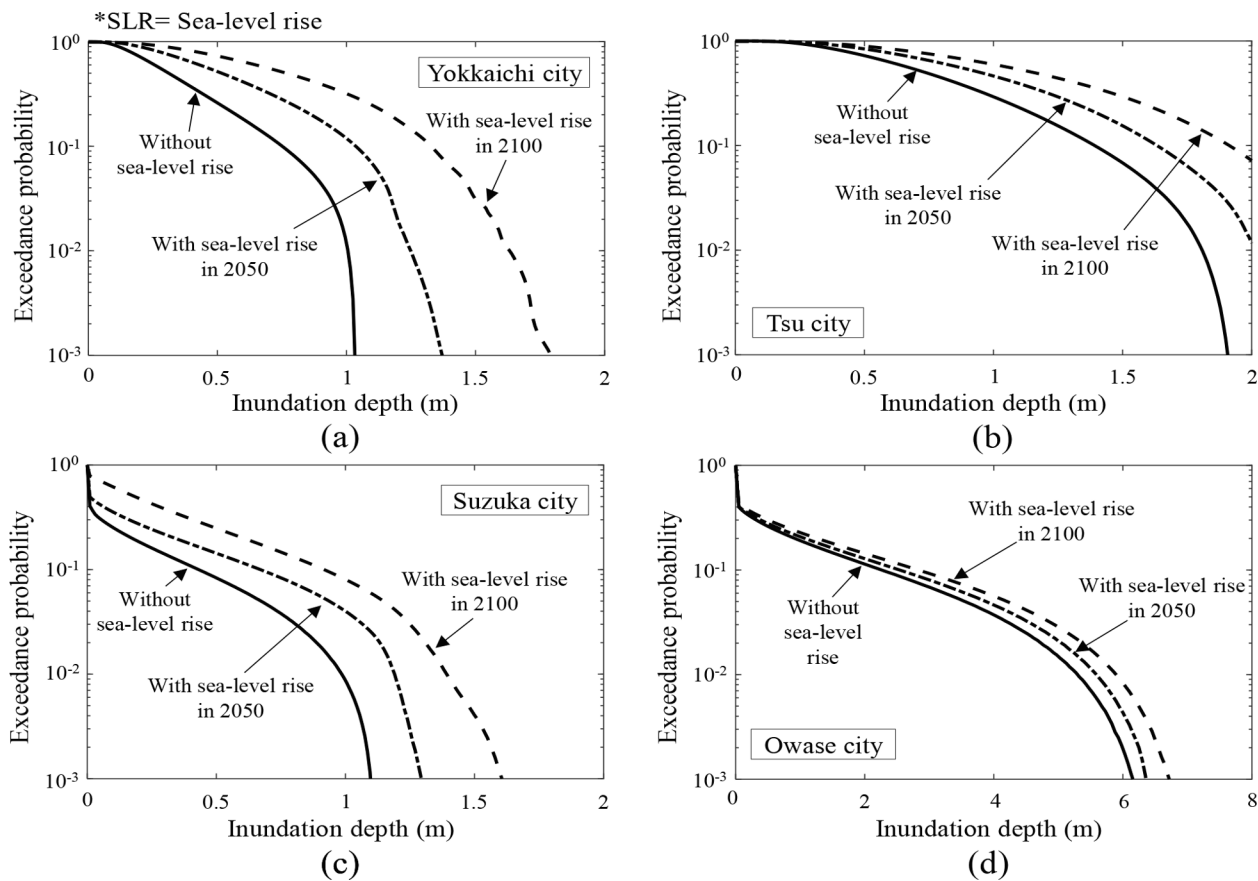


Fig. 16. Tsunami hazard curves considering the effects of sea-level rise in 2050 and 2100 for (a) Yokkaichi city, (b) Tsu city, (c) Suzuka city, and (d) Owase city.

Table 2

Sea-level rise effect on the tsunami inundation depth for various exceedance probabilities in 2100.

Tsunami Hazard	Inundation depth (m)											
	Yokkaichi city			Tsu city			Suzuka city			Owase city		
	Before SLR	After SLR	Ratio	Before SLR	After SLR	Ratio	Before SLR	After SLR	Ratio	Before SLR	After SLR	Ratio
95%	0.087	0.211	2.413	0.236	0.358	1.515	0.001	0.002	2.289	0.004	0.004	1.004
75%	0.185	0.477	2.576	0.474	0.753	1.591	0.004	0.027	6.446	0.021	0.021	1.017
50%	0.311	0.767	2.466	0.736	1.151	1.565	0.008	0.184	21.718	0.042	0.043	1.040
25%	0.511	1.095	2.143	1.067	1.577	1.478	0.103	0.497	4.829	0.563	0.821	1.459
5%	0.883	1.448	1.640	1.577	2.079	1.319	0.673	1.147	1.705	3.550	4.204	1.184

Note: SLR = Sea-level rise (SLR); and Ratio = After SLR/ Before SLR.

city due to the significantly higher tsunami inundation depth compared to the sea-level rise contribution. These findings are supported by the higher elevation of Owase city. Overall, the effects of the sea-level rise on the tsunami inundation depth become less notable as the land elevation increases. The effects of sea-level rise depends on the geographical conditions of the corresponding municipalities.

Figure 16 shows the tsunami hazard curves considering the effects of sea-level rise in several municipalities in Mie Prefecture, obtained by convolving the sea-level rise hazard curves shown in Fig. 11 and conditional tsunami hazard curves shown in Fig. 15 based on the total probability theorem in Eq. (1). Based on assumption [D] presented in Section 2, tsunami hazard curves are obtained considering that the Nankai-Tonankai earthquake will occur in 2050 or 2100. As shown in Fig. 16, the sea-level rise effects on tsunami hazards in 2050 are smaller than those in 2100. Moreover, considering all RCP scenarios, the tsunami hazards for all municipalities in the Hokusei region in 2100 increase between 0.4 and 0.6 m. According to this result, neglecting the

effects of sea-level rise in tsunami hazard assessments can lead to inappropriate prioritization of the disaster countermeasures among municipalities. In Owase city, due to severe hazards, the sea-level rise effect is insignificant.

Table 2 shows the tsunami hazard intensities and associated amplification factors due to sea-level rise effects. It can be concluded that when the sea-level rise hazard is considered, the inundation depth in the Hokusei region tends to be intensified with a scale between 1.5 and 2.5. In Owase city, the tsunami intensity slightly increases.

Based on the results obtained from the illustrative example, this paper provides evidence that sea-level rise can intensify the tsunami hazards depending on the geographical conditions of the analyzed area. Tsunamis with intensified impacts due to climate change can deteriorate the functionality of infrastructure systems and generate a more significant amount of disaster waste [18], which implicitly extends economic losses. By considering the effects of sea-level rise in tsunami hazard assessments, the risks to structures and infrastructure systems can be

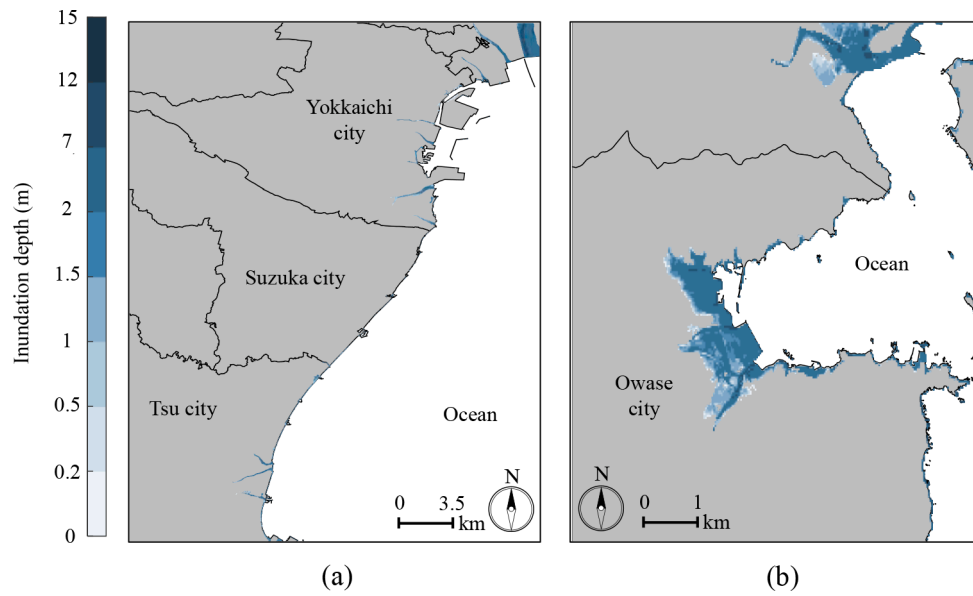


Fig. 17. Coastal defense performance in alleviating the effects of the tsunami and 1.2 m of sea-level rise for (a) northern Mie municipalities and (b) Owase city.

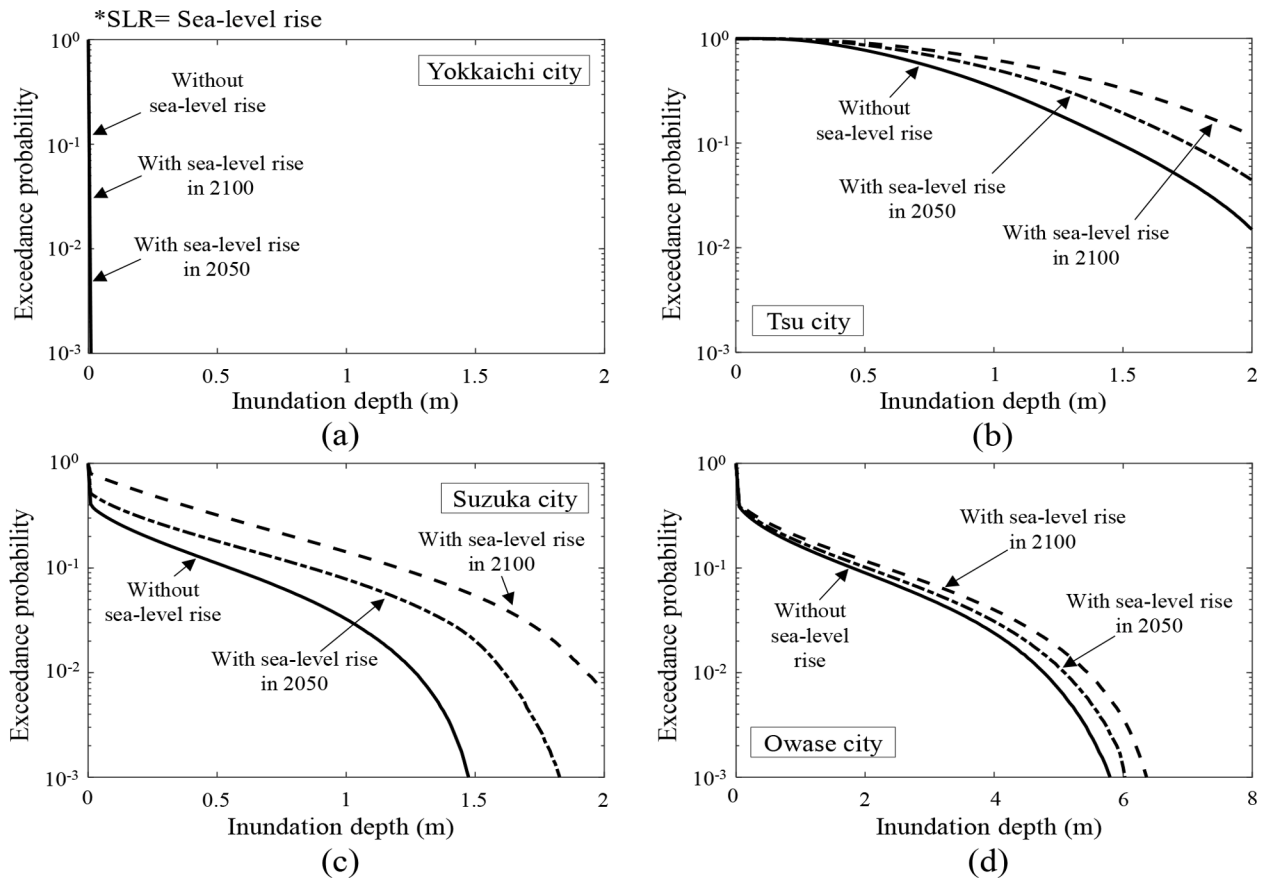


Fig. 18. Tsunami hazard curves considering the effects of sea-level rise and coastal defense in 2050 and 2100 for (a) Yokkaichi city, (b) Tsu city, (c) Suzuka city, and (d) Owase city.

more accurately evaluated.

5.3.4. Coastal defense influence in tsunami hazard assessment

To investigate the effects of the coastal defense on tsunamis and sea-level rise hazards, seawall and river embankment data for the Hokusei region and Owase city are considered. Fig. 17 shows the influence of the

seawall on the tsunami inundation depth in Mie prefecture given a sea-level rise of 1.2 m and fault parameter combination similar to that shown in Fig. 13. Assuming that coastal defenses have sufficient resistance against ground motion and tsunamis, the seawall can help alleviate the tsunami and sea-level rise impacts in Hokusei region. However, when the maximum wave height exceeds the seawall elevation, the

tsunami can propagate further to the land. This finding applies to Owase city, where the tsunami wave height is considerably larger than the seawall elevation.

Figure 18 presents the implications of the seawall on tsunami hazards in the same area shown in Fig. 14. By considering seawalls in the analysis, tsunami hazard evaluated in the land, such as in Yokkaichi city, can be significantly reduced. However, in Tsu and Suzuka city, where the analyzed grids are evaluated in the shoreline, the tsunami hazard increases due to the accumulating tsunami wave in front of the seawall. In Owase city, the presence of seawalls exerts a negligible effect due to the large tsunami wave height, and the tsunami hazard is expected to remain high. Hence, the performance of coastal defenses depends on the expected tsunami wave height in the affected areas and geographical conditions. Coastal defenses can help reduce the inundated area due to tsunamis and enhance the community resilience in coastal regions. However, in both the presence and absence of coastal structures, the sea-level rise tends to exacerbate the tsunami impacts and hazards. Therefore, local governments in Mie Prefecture should consider implementing and reinforcing the existing coastal defenses as a countermeasure against future tsunami disasters.

6. Conclusions

The following conclusions can be drawn:

- 1) A novel approach for probabilistic tsunami hazard assessment that encompasses sea-level rise uncertainties and their spatial variability is established. Integrated tsunami hazard curves that consider the effects of sea-level rise are generated by convolving the conditional tsunami hazard curves and corresponding sea-level rise hazard based on the total probability theorem.
- 2) A probabilistic regional sea-level rise hazard assessment is conducted considering the uncertainties associated with the sea-level rise prediction based on the available climate models under various climate change emission scenarios. The regional sea-level rise hazard map provided in this paper can be used as a reference when investigating increases in the tsunami hazard due to sea-level rise.
- 3) The tsunami intensity depends on the geographical conditions and distance between the analyzed location and earthquake fault. The inundation depth is expected to be higher as the analyzed location approaches the earthquake fault, while wider topographic plains can induce a more extensive tsunami inundation area.
- 4) The tsunami hazard intensity tends to increase linearly with the sea-level rise. The effects of the sea-level rise on tsunami hazards are primarily dependent on the land elevation. In addition, the tsunami inundation depth rapidly increases when the sea-level rise intensity is greater than the land elevation.
- 5) The tsunami hazard in the Hokusei region is expected to double in intensity due to the effects of sea-level rise. Neglecting the effects of sea-level rise in tsunami hazard assessments can lead to inadequate structural safety assessments against tsunamis and inappropriate prioritization of disaster countermeasures in the associated municipalities.

The proposed framework can be applied when performing risk and resilience assessments of infrastructure systems under tsunami hazards to consider the effects of future climate change. Three sea-level rise components are considered in this paper. Further research should include the contributions from additional sea-level rise components, such as land water storage and glacial isostatic adjustment, to assess the effects of sea-level rise more accurately on the tsunami hazard intensities. In addition, a more advanced fault model considering the extension of the earthquake rupture areas, coseismic deformation over land, plausible fault model scenarios, and varying tidal levels should be developed in future research to obtain more representative outcomes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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